

$$\begin{array}{l}
S^p_+\\
S^{d-1}_+\\
\mathcal{M}\subset R^D\\
\mathcal{D}\\
\mathcal{D}\ll \mathcal{D}\\
R^D\\
\mathcal{M}^b\\
R^b\\
\mathcal{M}^b\\
R^b\\
\mathcal{D}\in \mathcal{M}\\
T_\theta\mathcal{M}\subset R^D\\
\mathcal{D}\in P_\theta\subset R^{D\times D}\\
T_\theta\mathcal{M}\\
W_\beta\in R^{D\times d}\\
T_\theta\mathcal{M}\\
P_\theta=W_\theta W_\theta^\top\\
\psi_\theta:\mathcal{U}_\theta\rightarrow T_\theta\mathcal{M}\\
\psi_\theta(x)=\text{Proj}_{T_\theta\mathcal{M}}(x-\theta)\\
\mathcal{U}_\theta\\
\mathcal{M}\\
\phi_\theta=\psi_\theta^{-1}:T_\theta\mathcal{M}\rightarrow \mathcal{U}_\theta\\
\psi_\theta\in R^D\\
\mathcal{D}\in T_\theta\mathcal{M}\\
B\subset T_\theta\mathcal{M}\\
\psi_\theta^{-1}(B)\subset R^D\\
\mathcal{M}\\
C^{3,L}_{r,L}\\
\theta^*\\
\phi_\theta\\
L\\
T_{\theta^*}\mathcal{M}\\
\mathcal{M}\\
\ell:\\
\mathcal{X}^\times\rightarrow \mathcal{M}\\
R\\
\text{grad}_\theta\ell(x,\theta)\in T_\theta\mathcal{M}\\
\ell(x,\cdot)\\
R(\theta)=E_P[\ell(X,\theta)]\\
\text{grad }R(\theta)\\
\theta^*=\arg\min_{\theta\in\mathcal{M}}R(\theta)\\
\text{grad }R(\theta^*)=0\\
\mathcal{M}\\
\Pi\\
\mathcal{M}
\end{array}$$

$$\Pi(\mathrm{d}\theta \mid X^{(n)}) = \frac{\exp\big(-\sum_{i=1}^n \ell(X_i,\theta)\big)\,\Pi(\mathrm{d}\theta)}{\int_{\mathcal{M}} \exp\big(-\sum_{i=1}^n \ell(X_i,\theta')\big)\,\Pi(\mathrm{d}\theta')}.$$

$$\begin{array}{l}
(1)\\
\theta\stackrel{\ell}{=}\arg\min_{\theta\in S_\Pi}\frac{1}{n}\sum_{i=1}^n\ell(X_i,\theta)\\
\mathcal{H}_{\theta^*}\\
R(\cdot)\\
\theta^*\\
T_{\theta^*}\mathcal{M}\\
\Delta_{\theta^*}=E[g(X,\theta^*)g(X,\theta^*)^\top]\\
g\\
\theta^*-
\end{array}$$